Worksheet 1.1 — Average & Instantaneous Rate of Change

Grade 12 Calculus and Vectors (MCV4U) · Unit 1: Rates of Change and Limits

The Big Ideas

Average rate $= \dfrac{f(b)-f(a)}{b-a}$ (secant slope). Instantaneous rate (tangent slope) is *estimated* by plugging a tiny spacing h (e.g. $h = 0.001$) into the difference quotient $\dfrac{f(x+h)-f(x)}{h}$ — the smaller h, the better the estimate. (Limits make this exact in the next lesson.)

- Average rate over $[a,b]$: $\dfrac{f(b)-f(a)}{b-a}$.
- Difference quotient: $\dfrac{f(x+h)-f(x)}{h}$ — the secant slope for spacing h.
- Estimate the instantaneous rate by using a very small h; shrinking h further improves the estimate.

The Big Picture

Two kinds of rate

The **average rate of change** of f over $[a,b]$ is the secant slope $\dfrac{f(b)-f(a)}{b-a}$; the **instantaneous rate** is the slope at a single point.

Secant to tangent

As the interval shrinks toward a point, the secant line approaches the **tangent** line the geometric idea behind the derivative.

Estimating

Compute average rates over ever-smaller intervals to close in on the instantaneous rate at a point.

Worked Examples

Example 1 — Average rate (secant)

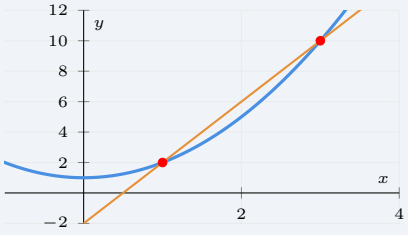
Average rate of change of $f(x) = x^2 + 1$ on $[1,3]$.

Solution. **Step 1 (evaluate the endpoints):** $f(1) = 1^2 + 1 = 2$ and $f(3) = 3^2 + 1 = 10$.

Step 2 (apply the formula): $\dfrac{f(3)-f(1)}{3-1} = \dfrac{10-2}{2}$.

Step 3 (simplify): $= \dfrac{8}{2} = 4$.

Conclusion: average rate $= 4$ — the secant through $(1,2)$ and $(3,10)$:



Example 2 — Average rate

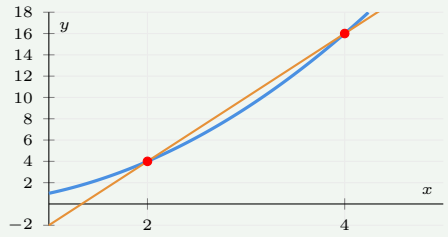
Average rate of change of $f(x) = x^2$ on $[2, 4]$.

Solution. **Step 1 (evaluate the endpoints):** $f(2) = 4$ and $f(4) = 16$.

Step 2 (apply the formula): $\dfrac{f(4) - f(2)}{4 - 2} = \dfrac{16 - 4}{2}$.

Step 3 (simplify): $= \dfrac{12}{2} = 6$.

Conclusion: average rate $= 6$ — the secant through $(2, 4)$ and $(4, 16)$:



Example 3 — Difference quotient

Simplify the difference quotient of $f(x) = x^2$ at a general x.

Solution. **Step 1 (form the quotient):** $\dfrac{f(x+h) - f(x)}{h} = \dfrac{(x+h)^2 - x^2}{h}$.

Step 2 (expand): $(x+h)^2 = x^2 + 2xh + h^2$, so the numerator is $2xh + h^2$.

Step 3 (factor and cancel h): $\dfrac{2xh + h^2}{h} = \dfrac{h(2x+h)}{h} = 2x + h$.

Conclusion: the difference quotient simplifies to $2x + h$.

Example 4 — Instantaneous rate — estimated

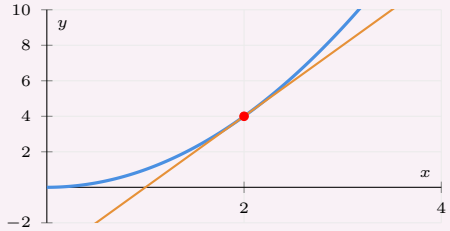
Estimate the instantaneous rate of $f(x) = x^2$ at $x = 2$ using $h = 0.001$.

Solution. **Step 1 (plug into the difference quotient):** $\dfrac{f(2.001) - f(2)}{0.001} = \dfrac{(2.001)^2 - 2^2}{0.001}$.

Step 2 (compute the squares): $(2.001)^2 = 4.004001$ and $2^2 = 4$.

Step 3 (subtract and divide): $\dfrac{4.004001 - 4}{0.001} = \dfrac{0.004001}{0.001} = 4.001$.

Conclusion: ≈ 4 — the near-tangent line at $x = 2$:



Example 5 — Instantaneous rate — estimated

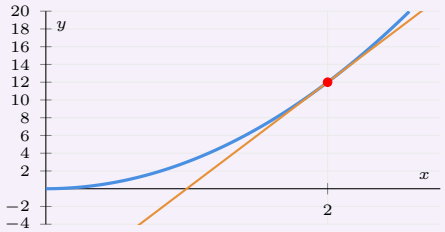
Estimate the instantaneous rate of $f(x) = 3x^2$ at $x = 2$ using $h = 0.001$.

Solution. **Step 1 (plug into the difference quotient):** $\dfrac{3(2.001)^2 - 3(2)^2}{0.001}$.

Step 2 (compute each term): $3(2.001)^2 = 3(4.004001) = 12.012003$ and $3(2)^2 = 12$.

Step 3 (subtract and divide): $\dfrac{12.012003 - 12}{0.001} = \dfrac{0.012003}{0.001} = 12.003$.

Conclusion: ≈ 12 — the near-tangent line at $x = 2$:



Example 6 — Velocity — estimated

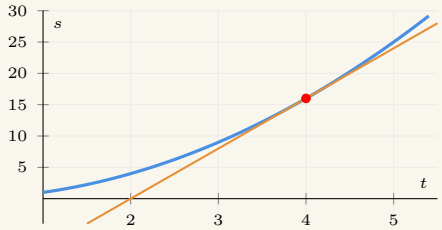
For $s(t) = t^2$, estimate the instantaneous velocity at $t = 4$ using $h = 0.001$.

Solution. **Step 1 (plug into the difference quotient):** $\dfrac{s(4.001) - s(4)}{0.001} = \dfrac{(4.001)^2 - 4^2}{0.001}$.

Step 2 (compute the squares): $(4.001)^2 = 16.008001$ and $4^2 = 16$.

Step 3 (subtract and divide): $\dfrac{16.008001 - 16}{0.001} = \dfrac{0.008001}{0.001} = 8.001$.

Conclusion: ≈ 8 — the near-tangent line at $t = 4$:



Example 7 — Average rate (cubic)

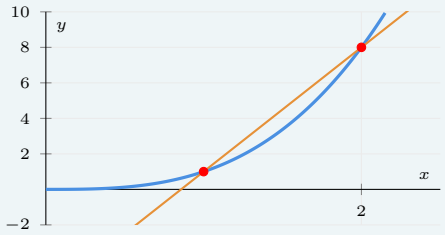
Average rate of change of $f(x) = x^3$ on $[1, 2]$.

Solution. **Step 1 (evaluate the endpoints):** $f(1) = 1^3 = 1$ and $f(2) = 2^3 = 8$.

Step 2 (apply the formula): $\dfrac{f(2) - f(1)}{2 - 1} = \dfrac{8 - 1}{1}$.

Step 3 (simplify): $= 7$.

Conclusion: average rate $= 7$ — the secant through $(1, 1)$ and $(2, 8)$:



Example 8 — Velocity from a model — estimated

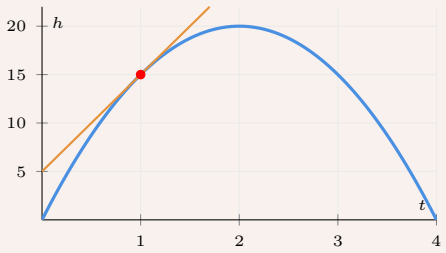
A ball's height is $h(t) = 20t - 5t^2$. Estimate its instantaneous velocity at $t = 1$ using $h = 0.001$.

Solution. **Step 1 (plug into the difference quotient):** $\dfrac{h(1.001) - h(1)}{0.001} = \dfrac{[20(1.001) - 5(1.001)^2] - [20(1) - 5(1)^2]}{0.001}$.

Step 2 (compute each height): $h(1.001) = 20.02 - 5.010005 = 15.009995$ and $h(1) = 20 - 5 = 15$.

Step 3 (subtract and divide): $\dfrac{15.009995 - 15}{0.001} = \dfrac{0.009995}{0.001} = 9.995$.

Conclusion: ≈ 10 m/s — the near-tangent line at $t = 1$:



Example 9 — Difference quotient

Simplify the difference quotient of $f(x) = 2x^2$ at $x = 1$.

Solution. **Step 1 (form the quotient):** $\dfrac{f(1 + h) - f(1)}{h} = \dfrac{2(1 + h)^2 - 2(1)^2}{h}$.

Step 2 (expand): $2(1+h)^2 = 2(1+2h+h^2) = 2+4h+2h^2$, so the numerator is $2+4h+2h^2-2 = 4h+2h^2$.

Step 3 (factor and cancel h): $\dfrac{4h + 2h^2}{h} = \dfrac{h(4 + 2h)}{h} = 4 + 2h$.

Conclusion: the difference quotient simplifies to $4 + 2h$.

Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

Average rate of change of $f(x) = x^2 - 3$ on $[2, 5]$?

Question 2

Average rate of change of $f(x) = x^2$ on $[0, 4]$?

Question 3

Simplify the difference quotient of $f(x) = x^2$ at $x = 3$.

Question 4

Estimate the instantaneous rate of $f(x) = x^2$ at $x = 5$ using $h = 0.001$?

Question 5

Estimate the instantaneous rate of $f(x) = 4x^2$ at $x = 1$ using $h = 0.001$?

Question 6

For $s(t) = t^2$, estimate the instantaneous velocity at $t = 6$ using $h = 0.001$?

Question 7

Average rate of change of $f(x) = x^3$ on $[0, 2]$?

Question 8

Simplify $\dfrac{3(2+h)^2 - 12}{h}$.

Question 9

A ball's height is $h(t) = 30t - 5t^2$; estimate its instantaneous velocity at $t = 2$ using $h = 0.001$?

Question 10

Average rate of change of $f(x) = 2x + 1$ on $[1, 4]$?

Question 11

Estimate the instantaneous rate of $f(x) = x^2$ at $x = -3$ using $h = 0.001$?

Question 12

For $s(t) = t^2 + t$, estimate the instantaneous velocity at $t = 1$ using $h = 0.001$?

Question 13 — Challenge

A population is $P(t) = 100 + t^2$; estimate its instantaneous rate of growth at $t = 5$ using $h = 0.001$.

Question Recap & Answer Key

Question Recap

1. Average rate of change of $f(x) = x^2 - 3$ on $[2, 5]$?
2. Average rate of change of $f(x) = x^2$ on $[0, 4]$?
3. Simplify the difference quotient of $f(x) = x^2$ at $x = 3$.
4. Estimate the instantaneous rate of $f(x) = x^2$ at $x = 5$ using $h = 0.001$?
5. Estimate the instantaneous rate of $f(x) = 4x^2$ at $x = 1$ using $h = 0.001$?
6. For $s(t) = t^2$, estimate the instantaneous velocity at $t = 6$ using $h = 0.001$?
7. Average rate of change of $f(x) = x^3$ on $[0, 2]$?
8. Simplify $\dfrac{3(2+h)^2 - 12}{h}$.
9. A ball's height is $h(t) = 30t - 5t^2$; estimate its instantaneous velocity at $t = 2$ using $h = 0.001$?
10. Average rate of change of $f(x) = 2x + 1$ on $[1, 4]$?
11. Estimate the instantaneous rate of $f(x) = x^2$ at $x = -3$ using $h = 0.001$?
12. For $s(t) = t^2 + t$, estimate the instantaneous velocity at $t = 1$ using $h = 0.001$?
13. A population is $P(t) = 100 + t^2$; estimate its instantaneous rate of growth at $t = 5$ using $h = 0.001$.

✓ Answer Key

1. $\frac{22-1}{3} = 7$
2. 4
3. $6 + h$
4. ≈ 10
5. ≈ 8
6. ≈ 12
7. $\frac{8-0}{2} = 4$
8. $12 + 3h$
9. ≈ 10 m/s
10. 2
11. ≈ -6
12. ≈ 3
13. ≈ 10